Solving systems of equations with the [redacted]ination method

1 Consider the following linear equations:

$$3x + 7y = -6$$
$$2x - y = -17$$

a What number should the second equation be multiplied by to eliminate y?

b Eliminate y and hence find the value of x that satisfies both equations.

c Substitute the x value to find the value of y that satisfies both equations.

2 Consider the following linear equations:

$$\frac{2x}{5} + \frac{3y}{5} = -\frac{7}{5}$$
$$-\frac{1}{4}\left(-5x + \frac{5y}{9}\right) = \frac{5}{9}$$

a Write the new equations after the fractional coefficients have been changed to integer coefficients and simplified.

b Eliminate y and hence find the value of x that satisfies both equations.

3 Consider the following linear equations:

$$-8x - y = 0$$
$$-5x + 3y = 6$$

To solve this system using the elimination method, first multiply both sides of the first equation by 3 and then add the resulting equation to the second equation.

a Write the new equation formed from the above procedure.

b Find the x and y values that satisfy both equations.

4 When we solve a system of equations using the elimination method, how do we know when there is no solution?

5 For each of the following systems of linear equations:

i Find the number of solutions that satisfy both equations.

ii State whether the system of equations is dependent, consistent or inconsistent.

a $\begin{aligned} 7x + 3y &= 4 \\ 14x + 6y &= 8 \end{aligned}$

b $\begin{aligned} 7x + 3y &= 5 \\ 28x + 12y &= 30 \end{aligned}$

6 Consider the following linear equations:

$$\begin{aligned} \frac{4x}{5} + \frac{3y}{5} &= 4 \\ 8x - 3y &= 5 \end{aligned}$$

State the operation required to change the fractional coefficients to integer coefficients in the first equation.

7 Explain how to eliminate the variable y in the following systems of equations:

a $\begin{aligned} \frac{1}{7}x + \frac{4}{7}y &= 7 \\ \frac{1}{2}x + \frac{1}{3}y &= 5 \end{aligned}$

b $\begin{aligned} 2x - 1.8y &= 5.9 \\ 0.65x + 0.07y &= -0.02 \end{aligned}$

8 Solve the following systems of equations:

a $\begin{aligned} 7x - 4y &= 15 \\ 7x + 5y &= 60 \end{aligned}$

b $\begin{aligned} -6x - 2y &= 46 \\ -30x - 6y &= 246 \end{aligned}$

c $\begin{aligned} -5x + 16y &= 82 \\ 25x - 4y &= 122 \end{aligned}$

d $\begin{aligned} x + \frac{5}{4}y &= \frac{9}{4} \\ \frac{3}{5}x + y &= \frac{7}{5} \end{aligned}$

e $\begin{aligned} -\frac{x}{4} + \frac{y}{5} &= 8 \\ \frac{x}{5} + \frac{y}{3} &= 1 \end{aligned}$

f $\begin{aligned} 5x + 3y &= 7 \\ x + y &= 2 \end{aligned}$

g $\begin{aligned} -5p - 7q &= -\frac{43}{5} \\ -15p - 28q &= -\frac{157}{5} \end{aligned}$

9 Solve the following system of equations:

$$\begin{aligned} 2x + 5y &= 9 \\ x + y &= 10 \end{aligned}$$

and find the value of each of the following expressions:

a $x + 4y$

b $7x + 10y$

c $3y$